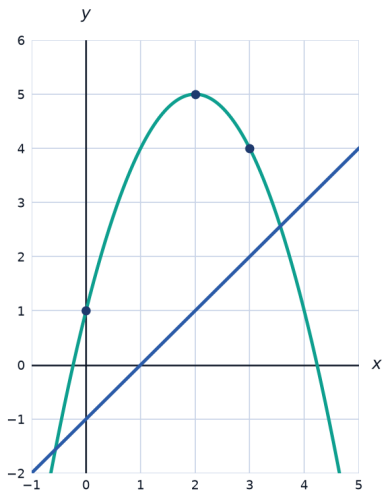
Exponential & Logarithmic Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Negative exponents

For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(5)$.

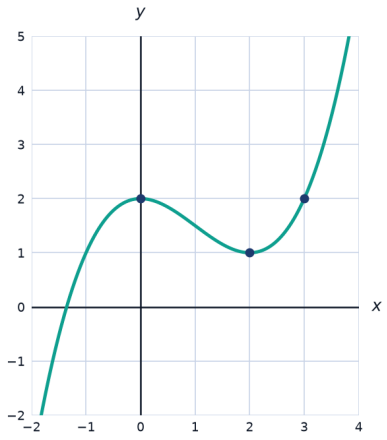


2 [Foundational] • Target: Negative exponents

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

3 [Foundational] • Target: The laws of exponents and scientific notation

For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(2)$.



4 [Foundational] • Target: The laws of exponents and scientific notation

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

5 [Foundational] • Target: The laws of exponents and scientific notation

A data set recorded for the laws of exponents and scientific notation consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

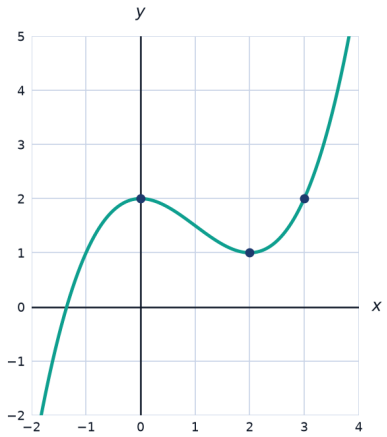
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6 [Foundational] • Target: The laws of exponents and scientific notation

For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(3)$.

7 [Foundational] • Target: The laws of exponents and scientific notation

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The laws of exponents and scientific notation

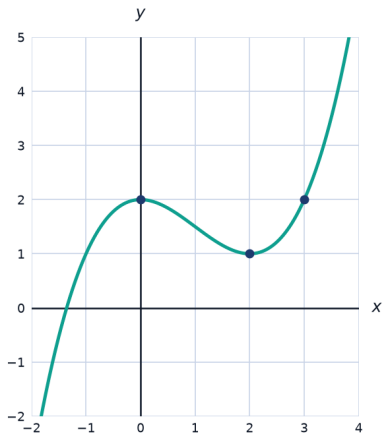
For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(4)$.

9 [Intermediate] • Target: The laws of exponents and scientific notation

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

10 [Intermediate] • Target: The laws of exponents and scientific notation

For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(5)$.



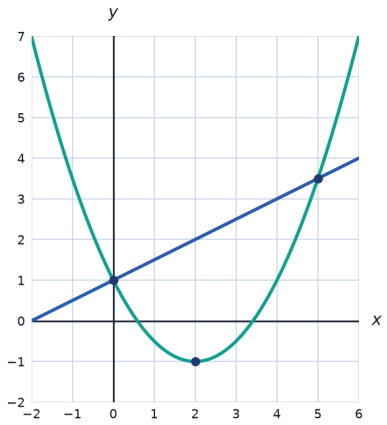
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11 [Intermediate] • Target: The laws of exponents and scientific notation

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

12 [Intermediate] • Target: Patterns with rational and irrational numbers

For the model $f(x) = 6x + 5$ in patterns with rational and irrational numbers, evaluate $f(2)$.



13 [Intermediate] • Target: Patterns with rational and irrational numbers

A data set recorded for patterns with rational and irrational numbers consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean? Show your complete algebraic work.

14 [Intermediate] • Target: Patterns with rational and irrational numbers

For the model $f(x) = 6x + 5$ in patterns with rational and irrational numbers, evaluate $f(3)$.

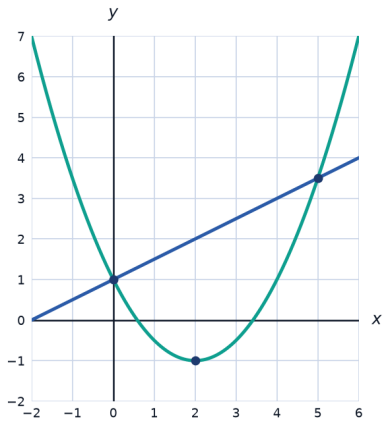
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Patterns with rational and irrational numbers • Cross-Lesson: 1.01 Linear Equations & Lines

For the model $f(x) = 6x + 5$ in patterns with rational and irrational numbers, evaluate $f(4)$.

16 [Challenging] • Target: Patterns with rational and irrational numbers • Cross-Lesson: 10.01 Rational Expressions

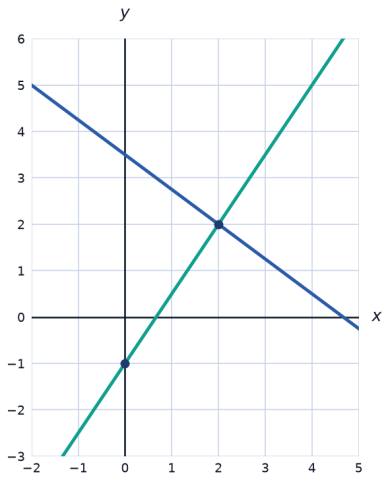
For the model $f(x) = 6x + 5$ in patterns with rational and irrational numbers, evaluate $f(5)$.



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17 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 11.01 Statistics & Data Analysis

For the model $f(x) = 2x + 6$ in simplifying radical expressions, evaluate $f(2)$.



18 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 2.01 Quadratic Functions

A data set recorded for simplifying radical expressions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

19 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 3.01 Exponents & Polynomials

For the model $f(x) = 2x + 6$ in simplifying radical expressions, evaluate $f(3)$.

20 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 4.01 Functions & Transformations

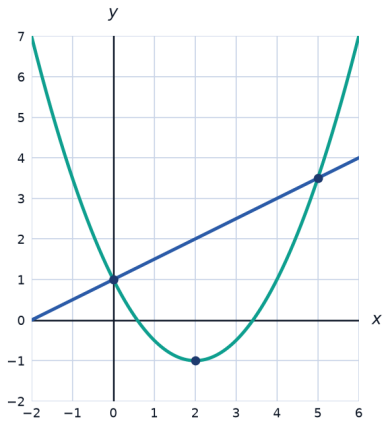
For the model $f(x) = 2x + 6$ in simplifying radical expressions, evaluate $f(4)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiplication properties of exponents

For the model $f(x) = 2x + 8$ in multiplication properties of exponents, evaluate $f(2)$.



a. 12

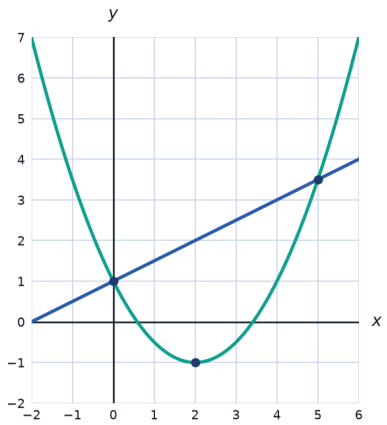
b. 14

c. 4

d. 16

22 [Foundational] • Target: Multiplication properties of exponents

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

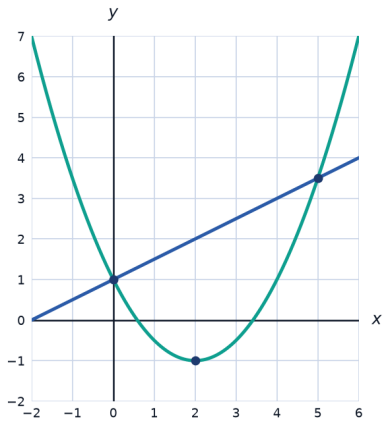
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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23 [Foundational] • Target: Multiplication properties of exponents

For the model $f(x) = 2x + 8$ in multiplication properties of exponents, evaluate $f(5)$.



a. 18

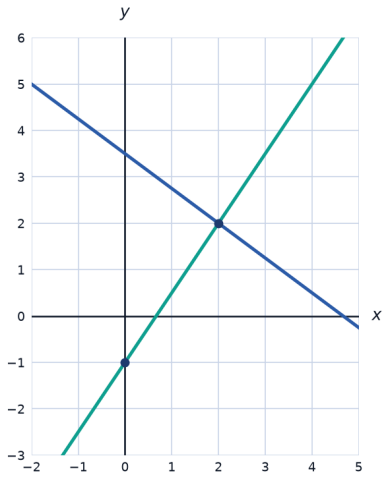
b. 20

c. 10

d. 22

24 [Foundational] • Target: Division properties of exponents and the zero exponent

For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(2)$.



a. 15

b. 18

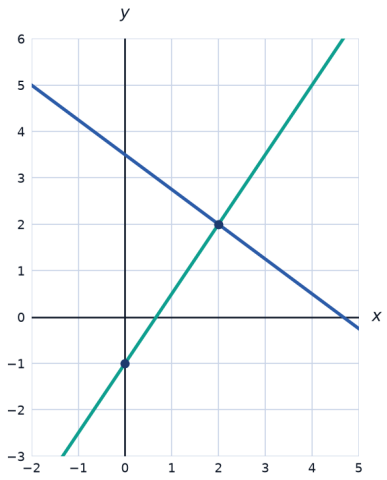
c. 6

d. 19

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25 [Foundational] • Target: Division properties of exponents and the zero exponent

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

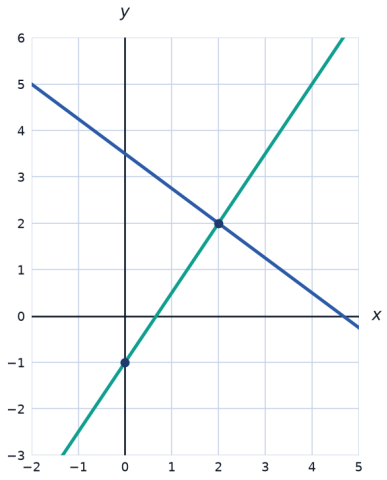
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

26 [Foundational] • Target: Division properties of exponents and the zero exponent

For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(5)$.



a. 24

b. 27

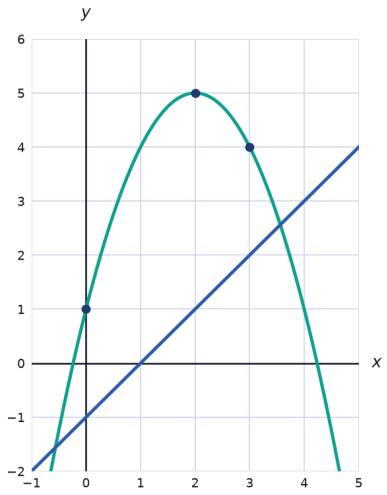
c. 15

d. 28

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27 [Foundational] • Target: Negative exponents

For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(2)$.



a. 11

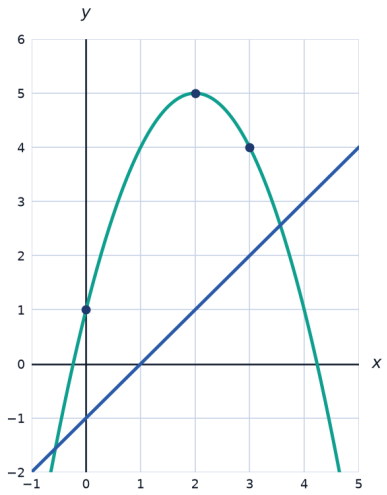
b. 15

c. 8

d. 15

28 [Foundational] • Target: Negative exponents

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiplication properties of exponents

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Multiplication properties of exponents

A data set recorded for multiplication properties of exponents consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

31 [Intermediate] • Target: Multiplication properties of exponents

For the model $f(x) = 2x + 8$ in multiplication properties of exponents, evaluate $f(3)$.

a. 14

b. 16

c. 6

d. 18

32 [Intermediate] • Target: Multiplication properties of exponents

For the model $f(x) = 2x + 8$ in multiplication properties of exponents, evaluate $f(4)$.

a. 16

b. 18

c. 8

d. 20

33 [Intermediate] • Target: Multiplication properties of exponents

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

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34 [Intermediate] • Target: Multiplication properties of exponents

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

35 [Intermediate] • Target: Division properties of exponents and the zero exponent

A data set recorded for division properties of exponents and the zero exponent consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

36 [Intermediate] • Target: Division properties of exponents and the zero exponent

For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(3)$.

a. 18

b. 21

c. 9

d. 22

37 [Intermediate] • Target: Division properties of exponents and the zero exponent

For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(4)$.

a. 21

b. 24

c. 12

d. 25

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Division properties of exponents and the zero exponent • Cross-Lesson: 2.01 Quadratic Functions

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

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39 [Challenging] • Target: Division properties of exponents and the zero exponent • Cross-Lesson: 3.01 Exponents & Polynomials

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

40 [Challenging] • Target: Negative exponents • Cross-Lesson: 4.01 Functions & Transformations

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

41 [Challenging] • Target: Negative exponents • Cross-Lesson: 6.01 Systems & Inequalities

For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(3)$.

a. 15

b. 19

c. 12

d. 19

42 [Challenging] • Target: Negative exponents • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(4)$.

a. 19

b. 23

c. 16

d. 23

43 [Challenging] • Target: Negative exponents • Cross-Lesson: 8.01 Factoring Polynomials

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

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44 [Challenging] • Target: The laws of exponents and scientific notation • Cross-Lesson: 9.01 Radicals & Exponents

A data set recorded for the laws of exponents and scientific notation consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

45 [Challenging] • Target: Patterns with rational and irrational numbers • Cross-Lesson: 1.01 Linear Equations & Lines

A data set recorded for patterns with rational and irrational numbers consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

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